Language Models for Automated Coding in Official Statistics: Risks and Opportunities

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Abstract

Statistical classifications, such as NACE, are especially important for National Statistical Offices (NSOs). Automating the classification process has been a focal point for NSOs in recent decades, and deterministic, rule-based methods that directly assign codes to natural language queries have been widely implemented. While accurate, these systems are labour-intensive and generalise poorly across domains, as each classification requires its own rule set. Recent advancements in natural language processing, particularly language modelling, offer an opportunity to improve the efficiency and scalability of the automatic coding process via semantic search systems. However, they come with their own set of challenges, most notably Uncertainty Quantification (UQ). We address UQ by employing robust statistical frameworks based on conformal prediction.

Keywords: Automatic Coding, Language Models, Conformal Prediction

1. Introduction

Large language models (LLMs) represent a recent and significant advancement in the field of Artificial Intelligence (AI), particularly in natural language processing (NLP). These models, usually based on Transformer architectures (Vaswani *et al.* 2017), are trained on large amounts of text to learn language patterns, semantic relationships, and contextual dependencies across various domains. Through this training, LLMs acquire the ability to understand and generate (in the case of auto-regressive transformers) text with impressive fluency and coherence.

Prominent examples, including the GPT series (Brown *et al.* 2020), LLaMA models (Touvron *et al.* 2023), and more recent reasoning models such as DeepSeek (Team 2024) and Qwen (Bai *et al.* 2024), demonstrate the increasing scale and capability of such systems, often involving hundreds of billions of parameters (Raffel *et al.* 2020). Unlike traditional machine learning models designed for narrowly defined predictive tasks, modern language models generalise across diverse linguistic applications, ranging from summarisation, translation, and question answering to text classification and representation. In the context of

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generative models, their capacity to produce contextually relevant and human-like responses positions them as valuable tools for enhancing information retrieval, automation, and decision support in data-intensive environments.

However, the probabilistic nature of LLMs poses challenges for interpretability, bias, and reliability. These models may reproduce social and linguistic biases from their training data and produce inaccurate or fabricated content, a phenomenon known as "hallucination" (Ji *et al.* 2023). Such issues highlight the importance of careful fine-tuning, human oversight, and ethical governance to ensure that LLMs are used responsibly, especially in public-sector settings such as official statistics.

Language models are not always generative. Text representation models, such as BERT (Devlin *et al.* 2019), S-BERT (Reimers and Gurevych 2019), and GTE (Li, Lin, *et al.* 2023), encode text into dense, high-dimensional vectors. This can be done at the token level (word or sub-word), where a sentence is represented as a variable-length sequence of dense vectors (token-level embeddings), or at the sentence level, where each sentence is represented as a single dense vector (sentence-level embedding). These representations are typically used for text classification and clustering tasks, where text generation is not required.

The introduction of LLMs presents significant opportunities for national statistical offices (Buono *et al.* 2024). Within statistical organisations, LLMs can automate routine administrative tasks such as drafting correspondence, writing reports, and preparing documentation, thereby improving operational efficiency (UNECE HLG-MOS 2025). This automation enables skilled personnel to focus on analytical and methodological tasks that require expert judgement and contextual understanding.

At the production level, LLMs can potentially be integrated into various phases of the Generic Statistical Business Process Model (GSBPM) (UNECE 2025). They can support questionnaire design, text classification, data validation, and the production of narrative interpretations of statistical outputs. Additionally, LLMs can facilitate metadata management, translate programming code between languages (e.g., SAS to R), and generate synthetic datasets for testing and methodological development. Conversational AI interfaces that use retrieval-augmented generation (RAG) can also transform data dissemination by allowing users to query statistical databases in natural language and receive context-specific explanations or visualisations.

However, these opportunities must be weighed against the associated risks. The opaque and data-dependent design of LLMs raises concerns about the accuracy, neutrality, and confidentiality of AI-generated outputs — principles that underpin official statistics (United Nations 2014). To protect these principles, human review and clear governance mechanisms are essential. Establishing robust ethical frameworks, transparency standards, and validation protocols is crucial to ensuring that the integration of LLMs enhances rather than undermines institutional credibility.

Finally, as public engagement with information increasingly occurs through AI-driven interfaces, statistical agencies face a strategic need to maintain their role as trusted sources of data. Collaborative partnerships among statistical organisations, academia, and AI developers can foster the development of reliable and verifiable generative AI applications that broaden access to official data while ensuring accuracy, accountability, and public trust (UNECE HLG-MOS 2023).

In this paper, we examine the use of LLMs to assign ATECO codes to economic activities

based on user-provided descriptions. The current tool used by Istat is CIRCE (Istat 2016), developed in R. It mainly relies on a thesaurus that has been manually updated year by year by dedicated staff. Users' descriptions are compared with the thesaurus content, and one or more ATECO codes are generated using string matching algorithms.

The transition to the new ATECO 2025 (Istat 2025) classification would require revising the thesaurus, which is both costly and time-consuming. LLMs could potentially provide a more flexible alternative for this transition.

In this work, we use two different kind of language models for different purposes: (1) *generative* LLMs to create a dataset of synthetic user queries labelled with ATECO codes, and (2) *non-generative* LLMs, in particular sentence encoders, to encode these texts into a high dimensional representation space where we can perform retrieval/classification based on user queries. Results obtained via language models are assessed against CIRCE-assigned labels; therefore, each performance measure is to be interpreted as concordance with CIRCE rather than plain accuracy. We explore two different retrieval/classification strategies to assign ATECO codes to natural language queries: *cosine-based retrieval*, typical of retrieval-augmented generation (RAG) applications (Lewis *et al.* 2020), and *linear discriminant analysis*. Moreover, we address the issue of uncertainty quantification with language models applying Conformal Prediction techniques (Vovk *et al.* 2005; Angelopoulos, Bates, *et al.* 2023).

This paper aims to assess the feasibility of using LLMs for ATECO classification and to determine whether they represent a viable alternative to the traditional approach. It is important to note that we focus on ATECO 2007 rev. 2022 (hereafter referred to as ATECO 2022) rather than ATECO 2025, as we have CIRCE-labelled data for ATECO 2022. Additionally, we focus on 2-digit codes, specifically ATECO *divisions*. However, given the flexibility of our proposed approach, this system can be easily extended to the ATECO 2025 classification and other statistical classifications with relatively low effort.

2. Data

This study relies on three complementary datasets that together provide a comprehensive view of economic activity descriptions and their corresponding ATECO codes. The first dataset contains the *official ATECO 2022 classification*, which defines the hierarchical structure and textual descriptors used for coding. The second dataset consists of *CIRCE-labelled queries*, representing real-world user inputs collected through Istat's online classification service. Finally, a *synthetic dataset* was generated using a large language model (`GPT-5-mini`) starting from the official classification. Table 2.1 summarises the main features of these datasets.

2.1 ATECO Official Classification (2007 rev. 2022)

The ATECO classification groups economic activities that are assumed to be similar, following the principles of completeness and mutual exclusivity (Istat 2022). Each category belongs to a single branch, ensuring an exhaustive and coherent representation of the national production system. The hierarchy comprises six levels — Sections (letters A–U), Divisions

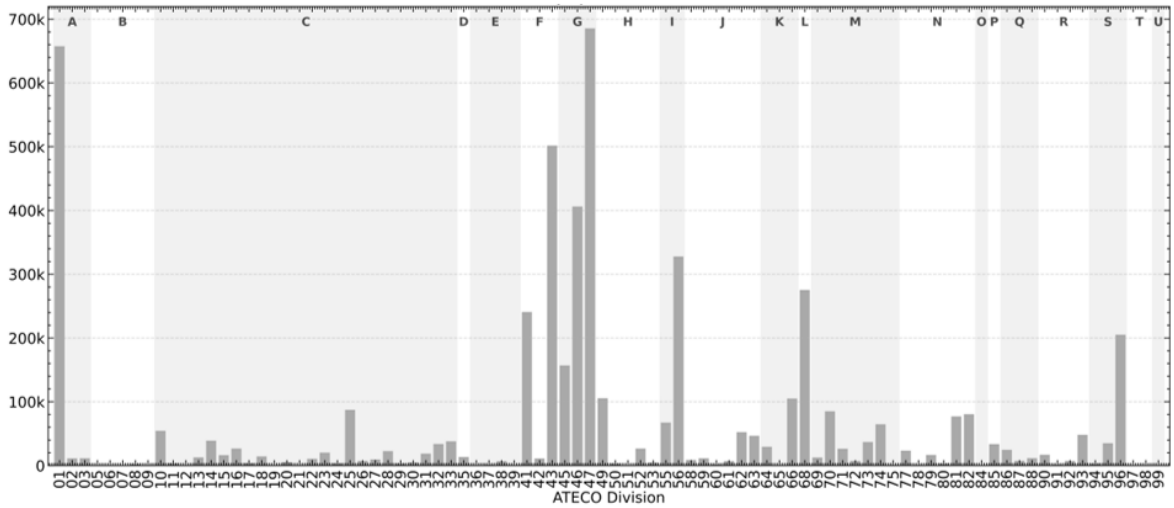
Table 2.1 – Summary of datasets used in the study.

Dataset	Source	Size / Scope	Purpose in study
ATECO 2022	Istat, official classification	21 Sections, 87 Divisions, ∼1,200 Sub-categories	Hierarchical reference and base for data augmentation
CIRCE-labelled queries	Istat, classification tool	33,544 user-generated queries	Benchmark for evaluating semantic vs. rule-based classification
Synthetic dataset	GPT-5-mini (OpenAI, 2025)	12,400 generated examples	Data augmentation, balancing and knowledge base creation

(2 digits), Groups (3 digits), Classes (4 digits), Categories (5 digits), and Sub-categories (6 digits) — covering 21 sections, 87 divisions, and more than 1,200 sub-categories in total.

Beyond numeric codes and titles, ATECO includes explanatory, inclusion, and exclusion notes that define the scope and boundaries of each code. General notes describe the overall domain of a branch; inclusion notes list typical activities covered by a code; and exclusion notes indicate those classified elsewhere. This detailed documentation supports consistency and comparability across statistical domains but also increases the linguistic complexity of the classification, posing challenges for automatic coding systems.

Figure 2.1 - Distribution of active businesses in Italy by ATECO division (2-digit code) as of December 2024. Shaded background areas and capital letters indicate ATECO sections.



Source: Own computation based on Istat Business Register.

Figure 2.1 shows the distribution of active businesses in Italy by ATECO division (2-digit level) as of December 2024. The figure highlights a highly unbalanced structure: for instance, Division 47 (Retail trade) accounts for more than 680,000 enterprises, whereas Division 99 (Activities of extraterritorial organisations and bodies) includes only a few cases. This imbalance has direct implications for the design of classification models, motivating the

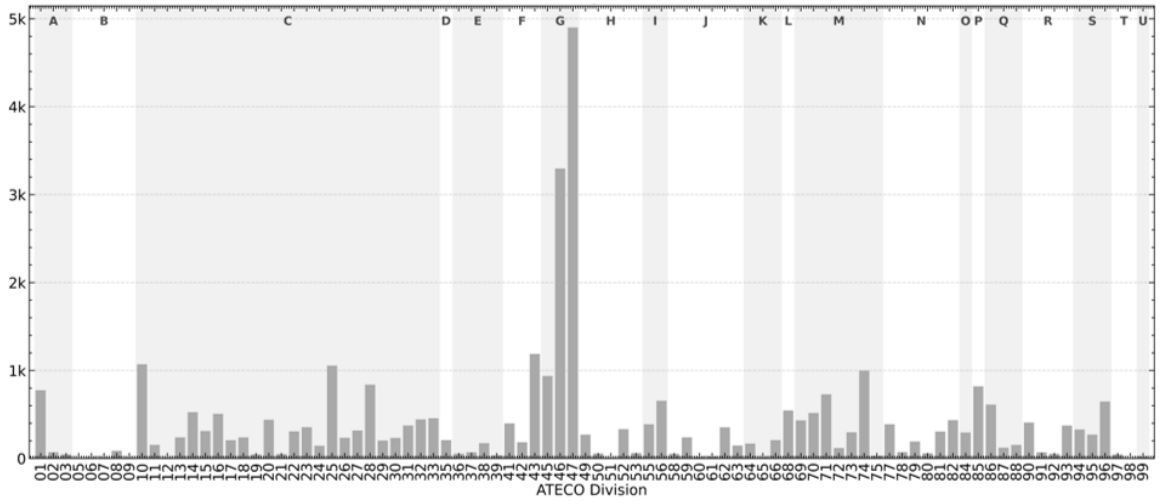
use of complementary data sources to ensure adequate representation across the entire taxonomy. Although this study focuses on the 2022 version, the proposed methods are designed to support the transition to ATECO 2025 (Istat 2025).

2.2 CIRCE-labelled Real-world Queries

To evaluate the proposed semantic-search approach, we rely on real-world data collected through CIRCE, the rule-based system currently used by the Italian National Institute of Statistics (Istat) to assist businesses and professionals in identifying the appropriate ATECO category from short natural-language descriptions (Istat 2016). CIRCE has been publicly available since 2016 and is accessible through the Istat classification portal.[2]

The dataset consists of 33,544 user queries, each processed through the CIRCE algorithm to obtain a predicted 5-digit ATECO code. This output constitutes a labelled corpus that reflects authentic user interactions with the system. The corpus is used as a benchmark to assess the alignment between traditional rule-based classification and embedding-based Semantic Search.

Figure 2.2 - Distribution of CIRCE-assigned labels across ATECO divisions (2-digit codes). Shaded background areas and capital letters indicate ATECO sections.



Source: Own computation based on CIRCE system outputs (Istat).

Figure 2.2 shows the distribution of CIRCE-assigned labels by ATECO division (2-digit level). The shape of this distribution broadly mirrors the overall population of Italian enterprises (see Figure 2.1), with Division 47 dominating the dataset and low-frequency divisions, such as Division 99, appearing only marginally. This imbalance is a key factor considered in the experimental design described in the following section.

[2] https://www.istat.it/classificazione/classificazione-delle-attivita-economiche-ateco-2007-aggiornamento-2022/

2.3 Synthetic Dataset

Sentence encoder models are trained to encode semantically similar sentences close in the sentence representation space. These models process large amounts of data during training and may contain inherent encoding biases. For instance, sentences that are similar in thematic content but differ in length, style, or overall tone can be projected into quite distant regions of the representation space. For this reason, it is ideal that, when classifying a natural language input query, the reference sentences (representative of the ATECO divisions) are somewhat similar to the input query. In theory, we could directly encode the official descriptions of ATECO codes, but this yields empirically lower performance. Moreover, some ATECO divisions include only a few codes, leading to a significant imbalance in the dataset.

For these reasons, we generated a synthetic dataset of business activity descriptions using a generative LLM, `GPT-5-mini` (OpenAI 2025). The model was instructed to create ten short queries for each ATECO sub-category (6-digit level), resulting in a total of 12,400 synthetic examples. Each prompt combined the official ATECO descriptors (including titles, explanatory notes, and inclusion/exclusion rules) to ensure semantic coherence with the taxonomy. Moreover, the model was given several real-world queries as examples to better reproduce their style and tone.

For each sub-category, the model generated realistic free-text descriptions of economic activities, mirroring the type of input provided by enterprises and professionals through the CIRCE platform. These synthetic queries were then grouped by the 2-digit division level.

3. Methodology

In this section, we will first briefly outline the idea behind CIRCE, then move onto describing Sentence Embeddings, describing different ways to use them for semantic similarity search, and finally show a way to ground our predictions in statistical frameworks.

3.1 CIRCE

Traditional automatic coding systems in official statistics rely on deterministic rule sets and weighted keyword matching. These approaches use dictionaries derived from the classification, where entries and user queries are pre-processed through tokenisation, stop-word removal, and synonym replacement, then represented as sets of weighted n-grams. Systems such as CIRCE (ACTR v3 engine) compare parsed queries with all dictionary entries and compute similarity scores to determine whether a match is unique, multiple, possible, or failed, based on score thresholds. While transparent and easily tuned by domain experts, these methods face key limitations: they lack semantic understanding, depend on constant maintenance of tokenisation and synonym rules, enforce rigid token-level matching, and scale poorly to new taxonomies. These issues, made evident during Italy's transition from ATECO 2022 to ATECO 2025, underscore the need for more flexible, language model–based approaches that capture meaning and context beyond surface-level tokens.

3.2 Sentence Embeddings

Token-level embeddings assign vectors to individual words or subwords, capturing local lexical and morphological information. Sentence-level embeddings, by contrast, compress the whole text into a single vector that captures global semantics and context, i.e., the overall meaning or intent of the sentence rather than the meaning of each token in isolation. For our task (classifying economic activities), this global view is often more useful, treating an activity description as a single semantic unit rather than a bag of token features.

$$e_s = \pi(s), \quad e_s \in \mathbb{R}^d$$

Our approach builds a representation space for the economic activity taxonomy, the *knowledge base*, in which each activity code is mapped to a dense, high-dimensional vector. We create these embeddings using sentence-level models (Reimers and Gurevych 2019), which produce a single fixed-length vector for an entire piece of text. This lets each code be represented as a compact semantic fingerprint in the shared vector space. In particular, we rely on the pre-trained `gte-multilingual-base` embedding model (Li, Lin, *et al.* 2023), which maps sentences to $768-$dimensional embedding vectors.

Since we have multiple sentences assigned to each ATECO division, we end up with a one-to-many (division-to-vectors) mapping.

Retrieval strategies Once activity codes are embedded in the shared vector space, semantic similarity between them can be efficiently measured using *cosine similarity*. This metric evaluates the cosine of the angle between two embedding vectors, effectively capturing how close their meanings are, regardless of vector magnitude. In this framework, retrieval becomes a geometric search problem: identifying the codes whose embeddings are most similar (i.e., closest in angle) to a given query vector, enabling intuitive, meaning-based lookup and matching across the taxonomy. Considering our one-to-many mapping, we define the similarity scores assigned to each ATECO division D_j, for input query x_i, as:

$$\text{score}(D_j) = \max\left(sim(\pi(x_i), e_{1,j})), \ldots, sim(\pi(x_i), e_{M,j})\right)$$

where $\pi(x_i)$ is the embedding vector assigned to query x_i, and $\{e_n\}_{i=1}^{M}$ are the M embedding vectors (corresponding to M sentences) that characterise division D_j. That is, each division is assigned the highest similarity score between all scores extracted from the corresponding vectors.

Another alternative approach we explore is Linear Discriminant Analysis (LDA) to retrieve relevant codes from the knowledge base. The LDA model is fit on the labelled Sentence Embeddings, and the probabilities of each ATECO division are retrieved.

3.3 Conformalising Semantic Search

To obtain valid and interpretable uncertainty estimates for semantic classification of economic activities, we employ Conformal Prediction (CP) procedures. Conformal prediction provides a distribution-free framework for constructing prediction sets, i.e., collections of

candidate labels that contain the true class with a user-specified probability $1 - \alpha$. We will explore both direct CP implementations, such as Split Conformal Prediction and Adaptive Prediction Sets, and Conformal Risk Control, which extends CP to any monotone loss function (Angelopoulos, Bates, *et al.* 2024).

3.3.1 Split Conformal Prediction and Adaptive Prediction Sets

Given a nonconformity score (or "uncertainty") score function $s(x, y)$ that quantifies how atypical a label y is given an input x, the conformal algorithm calibrates a threshold on these scores using a held-out calibration set. For a new query x_{new}, the conformal predictor then outputs all labels y whose scores do not exceed this calibrated threshold, hence guaranteeing coverage under exchangeability assumptions. While CP guarantees *marginal* coverage, it cannot guarantee *conditional* coverage in any of its forms. However, if best practices are followed, one can get close to it in practice (Angelopoulos, Bates, *et al.* 2023).

Formally, let (x, y) denote a random query and its true class, with $y \in \{1, \ldots, N\}$. We assume we have an embedding map $x \mapsto e(x) \in \mathbb{R}^d$ (our sentence encoder), and a finite calibration set $\mathcal{D}_{\text{cal}} = \{(x_i, y_i)\}_{i=1}^n$ drawn exchangeably with the test query. The goal is to turn a score function $s(x, y)$ (low score = plausible label) into a set-valued prediction $C_\alpha \subseteq \{1, \ldots, N\}$ that satisfies the usual marginal coverage guarantee:

$$\Pr\left(y \in C_\alpha(x)\right) \geq 1 - \alpha$$

for a user-chosen miscoverage level $\alpha \in (0, 1)$.

In our setting, each query is a natural-language description of an economic activity, and each possible class corresponds to an ATECO division (N pre-defined labels). Recalling previous paragraphs, we can define two alternative heuristics for the nonconformity score:

1. Cosine similarity, measuring semantic alignment between the query and class embeddings (lower similarity implies higher nonconformity);

2. Calibrated posterior probability from the Linear Discriminant Analysis (LDA) classifier, trained in the same embedding space, where uncertainty (or nonconformity) is represented as $1 - p(y|x)$.

As for our *calibration* $\mathcal{D}_{cal}$ and *validation* $\mathcal{D}_{val}$ sets, used for calibrating the procedure and computing coverage and diagnostics, respectively, we split the CIRCE-labelled dataset in half, leading to $16,772$ observations for each set. Note that the CIRCE dataset is never part of either the knowledge base creation or training processes.

The quality of CP procedures, beyond the coverage guarantee, is usually assessed via two properties: *efficiency* and *adaptivity*. The former depends on the overall size of prediction sets, where smaller sizes are associated with higher efficiency, while the latter refers to the procedure's ability to create larger prediction sets for harder examples, and smaller prediction sets for easier examples. Different CP algorithms lead to different efficiency and adaptivity properties. We apply both Split Conformal Prediction (SCP) and Adaptive Prediction Sets (APS) procedures (Lei *et al.* 2018; Romano *et al.* 2020).

Split Conformal Prediction Define a nonconformity score $q_i = s(x_i, y_i)$ for each calibration pair in $\mathcal{D}_{\text{cal}}$. Sort the n calibration scores in increasing order and let $q_{(k)}$ denote the $k-$th smallest.

For finite-sample marginal coverage, we use the quantile index

$$k^* = \frac{\lceil (n+1)(1-\alpha) \rceil}{n}, \quad \tau = q_{(k^*)}.$$

The Split Conformal Prediction set for a new query x_{new} is

$$C_\alpha(x_{new}) = \{y \in \{1, \ldots, N\} : s(x_{new}, y) \leq \tau\}$$

Under the exchangeability assumption, this guarantees

$$\Pr(y \in C_\alpha(x)) \geq 1 - \alpha.$$

The efficiency of this procedure can be measured by the average set size

$$\text{AvgSize}(\alpha) = \mathbb{E}[|C_\alpha(x)|].$$

Adaptive Prediction Sets (APS) When the model provides a probability distribution $\{p_k(x)\}_{k=1}^N$ over labels, APS constructs Adaptive Prediction Sets that reflect the model's uncertainty. For each calibration example, define the APS score for the true label y as the cumulative probability mass up to and including y, when classes are sorted in decreasing order of $p_k(x)$:

$$S(x, y) = \sum_{k: p_k(x) \geq p_y(x)} p_k(x).$$

Intuitively, $S(x, y)$ is the total probability mass of all labels ranked above (and including) the true one: small S means y lies among the top-probability labels.
Compute calibration scores $S_i = S(x_i, y_i)$ on $\mathcal{D}_{\text{cal}}$, then take the

$$\tau_{\text{APS}} = S_{(k^*)}, \quad k^* = \lceil (n+1)(1-\alpha) \rceil.$$

For a new input x, the APS prediction set is

$$C^\alpha_{\text{APS}}(x) = \{y : S(x, y) \leq \tau_{\text{APS}}\}.$$

Equivalently, $C^{\text{APS}}_\alpha(x)$ is the smallest set of top-ranked labels whose cumulative probability mass is at most τ_{APS}.
APS typically yields smaller, more informative sets on confident examples (where p is concentrated) and larger sets on ambiguous ones, while maintaining coverage at level $1 - \alpha$.

3.3.2 Conformal Risk Control

We also consider a Conformal Risk Control formulation to control the false negative rate (FNR) rather than the usual marginal miscoverage. In this setting, the goal is to produce set-valued predictions $C(x) \subseteq \{1, \ldots, N\}$ such that, with high confidence, the population FNR (probability that the true label is not included in the returned set) does not exceed a user-specified target α. Concretely, let $s(x, y)$ be a conformity score where larger values indicate greater conformity (e.g., cosine similarity or a calibrated class probability). For a threshold τ the set predictor is:

$$C_\tau(x) = \{y : s(x, y) \geq \tau\}$$

Using a held-out calibration set of size n, we compute the empirical FNR at candidate thresholds as:

$$F\hat{N}R(\tau) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{s(x_i, y_i) < \tau\},$$

and apply a one-sided concentration correction $\varepsilon = \sqrt{\log(1/\delta)/(2n)}$ for confidence level $1 - \delta$. To guarantee (with probability $\geq 1 - \delta$) that the true FNR is at most α, we select the largest threshold τ satisfying $F\hat{N}R(\tau) \leq \alpha - \varepsilon$ (if $\alpha - \varepsilon < 0$ we require $F\hat{N}R(\tau) = 0$). This procedure produces the smallest prediction sets that still satisfy the risk control guarantee. The method, like CP, only requires data exchangeability and can be applied with any monotone conformity score.

4. Results

In this section, we first present the results obtained using cosine similarity-based retrieval of correct labels and LDA-based label prediction. Then, we apply Conformal Prediction algorithms to both cosine similarity values and LDA output probabilities to build statistically valid prediction sets.

Table 4.1 – Performance of different retrieval strategies (matching with CIRCE labels).

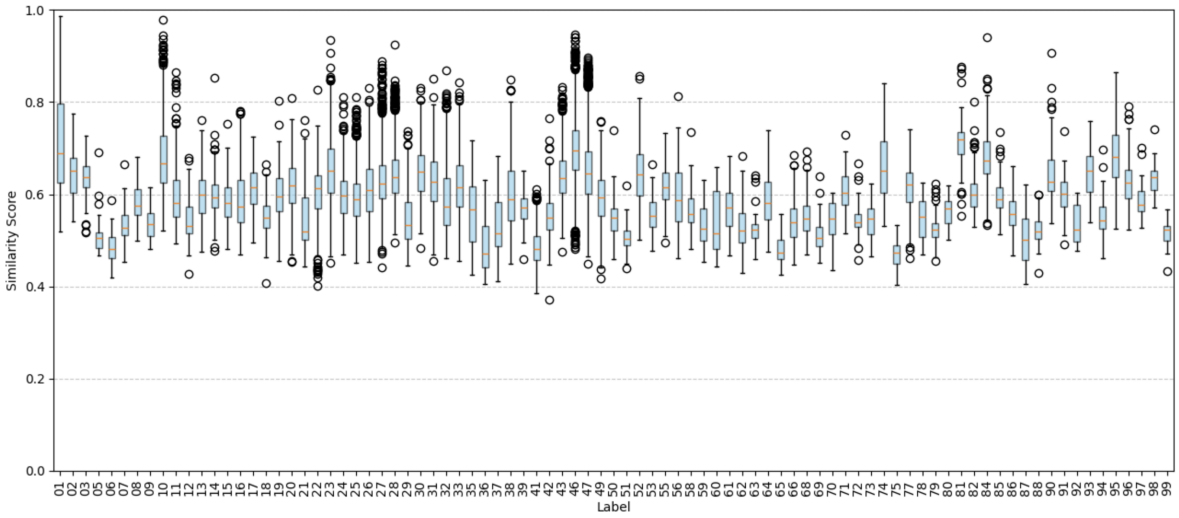
Strategy	Hit@1	Hit@3	Hit@5
Cosine similarity	0.549	0.827	**0.902**
LDA probabilities	**0.630**	**0.831**	0.881

Table 4.1 shows hit@1, hit@3, and hit@5 metrics. These are computed by checking whether the correct label is found inside the top_k extracted labels. We can see how LDA-based produces better results (i.e., more in line with CIRCE labels) at the first retrieved positions, while cosine similarity-based retrieval achieves higher performance when considering the first five extracted labels. However, producing fixed-size prediction sets, i.e., top 5 results, is not optimal for such applications. Therefore, we proceed to calibrate threshold values to create dynamically sized prediction sets.

4.1 Experiment A. Cosine Similarity as Conformity Score

In this first experiment, we use pair-wise cosine similarity scores, computed between each possible pair of query-code embedding vectors, as heuristic notions of uncertainty. In particular, lower cosine similarity scores indicate higher levels of nonconformity, while the contrary is true for higher cosine similarity values. Since we have a one-to-many mapping in our knowledge base, i.e., multiple embedding vectors represent the same ATECO division, we take the maximum cosine similarity for each division. This way, for each query, we end up with a score for each ATECO division. Figure 4.1 shows the distribution of cosine similarity values assigned to true labels (CIRCE-assigned labels) for each division. We can observe how cosine similarity scores can differ significantly across divisions. This is due to the pre-training of the language model (sentence encoder) and its inherent biases.

Figure 4.1 - Distribution of cosine similarity scores assigned to correct labels across ATECO divisions.

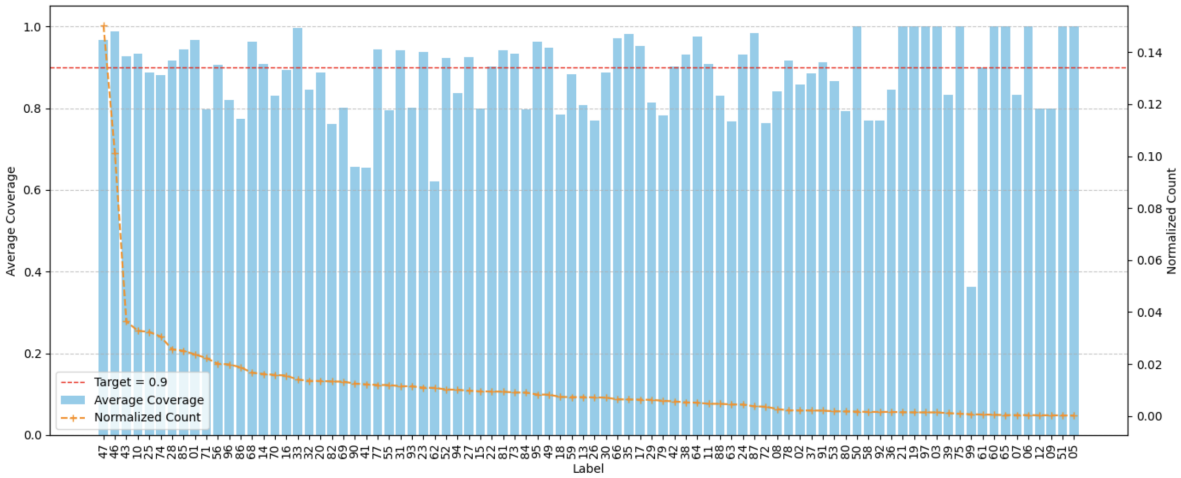


4.1.1 Standard Split CP Algorithm

We first apply a standard Split Conformal Prediction algorithm, one of the simplest CP applications. In this case, the nonconformity score is computed simply as $1-cos_sim(y_{true})$. Using these scores, we extract a calibrated 90th percentile $\hat{q}_{SCP}$ of 0.313, which we use to construct the prediction sets: each observation with a nonconformity score lower than $\hat{q}_{SCP}$ is included in the prediction set. This yields an average empirical coverage of 89.9%.

To inspect the label-stratified coverage (LSC) of our procedure, we plot the average coverage for each ATECO division (Figure 4.2). We order the divisions by their prevalence in the validation set, as we give more importance to more populated divisions. As for prediction set

Figure 4.2 - Average coverage for the split CP procedure (left axis) by ATECO division, ordered by normalised count (right axis). Cosine similarities as conformity scores



size, the average extracted set size is 8.09. This means that, on average, a query is assigned $\sim 9\%$ of the whole label space, which is a fairly large share.

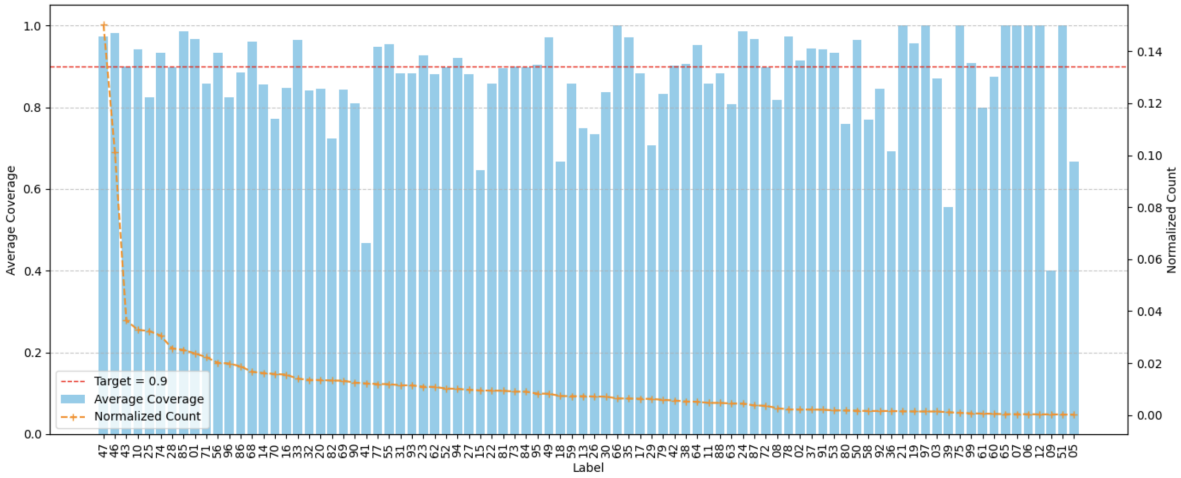
4.1.2 Adaptive Prediction Sets (APS)

Here, we use a different CP procedure, namely Adaptive Prediction Sets (APS). This time, the score function considers the entire cumulative distribution of cosine similarity values up to the correct label. In general, this should improve the procedure's adaptivity, i.e., its ability to adjust the set size to the complexity of the query and produce smaller sets on average. The extracted 90th calibrated percentile $\hat{q}_{APS}$ is 14.04: labels are included in the prediction set up until the cumulative distribution of their ordered cosine similarity scores reaches $1 - \hat{q}_{APS}$, yielding an average empirical coverage of 90.1%. Figure 4.3 shows the label-stratified coverage across ATECO divisions. Indeed, the average prediction set size for the APS procedure is 4.98, much lower than with Split Conformal Prediction, making the procedure substantially much more efficient.

4.1.3 Conformal Risk Control

Finally, we apply a Conformal Risk Control procedure using cosine similarity scores as conformity measures. The Conformal Risk Control attains the target FNR with the calibrated threshold, with an empirical coverage on the validation set of 0.909 (achieved FNR $= 0.091$) and $\tau = 0.682$. The average set size is fairly large (8.76 labels) and a small fraction of queries yield empty sets, which in practice would require a fallback. The label-stratified distribution of coverage (Figure 4.4) closely mimics the one obtained with Split Conformal Prediction, especially for the most frequent labels.

Figure 4.3 - Average coverage for the APS procedure (left axis) by ATECO division, ordered by normalised count (right axis). Cosine similarities as conformity scores.



4.2 Experiment B. LDA Probabilities as Conformity Scores

In this second experiment, we fit a linear discriminant analysis (LDA) model on the ATECO divisions' knowledge base. Each embedding dimension represents a feature ($n_{features} = 768$) and the GPT-generated dataset is split into a training and a test set, with a $80 : 20$ ratio. The output probabilities are calibrated using isotonic regression. We experimented with dimensionality reduction methods, such as PCA and UMAP, but they negatively impacted performance. We keep the same experimental design as in paragraph 4.1, utilising calibrated output probabilities as heuristic notions of uncertainty in place of cosine similarity scores.
Figure 4.5 shows the label-stratified distribution of the probabilities assigned to the correct ATECO divisions by the LDA model. As we can see, the values greatly vary both between and within labels.

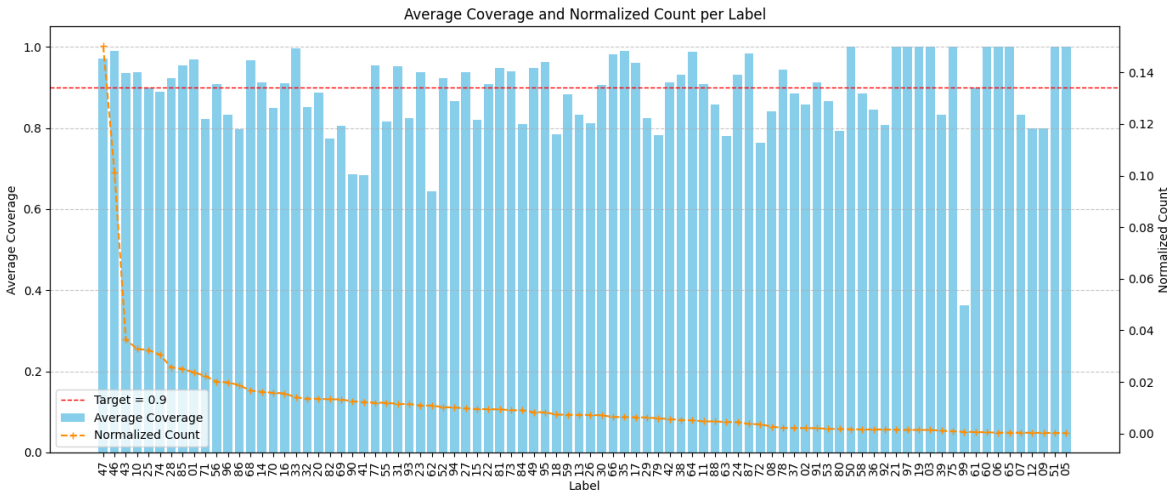
4.2.1 Standard Split CP Algorithm

Mirroring the previous paragraphs, the nonconformity scores for the SCP algorithm are computed as $1 - \hat{p}(y_{true}|x)$. We extract a 90% calibrated percentile $\hat{q}^{LDA}_{SCP} = 0.984$, meaning that labels that are assigned a probability higher than $1 - \hat{q}^{LDA}_{SCP}$. This yields an average empirical coverage of 90.1% prediction sets with an average size of 5.2, which is close to APS on cosine similarity scores. The label stratified coverage is shown in Figure 4.6.

4.2.2 Adaptive Prediction Sets (APS)

We can now apply an APS procedure using the LDA output probabilities. The extracted calibrated 90% percentile is 0.977, yielding an average empirical coverage of 90.2% and an

Figure 4.4 - Average coverage for the Conformal Risk Control procedure (left axis) by ATECO division, ordered by normalised count (right axis). Cosine similarities as conformity scores.



average set size of 7.36, which is closer to the standard CP procedure using cosine similarity scores. Figure 4.8 shows the label-stratified coverage for the APS procedure.

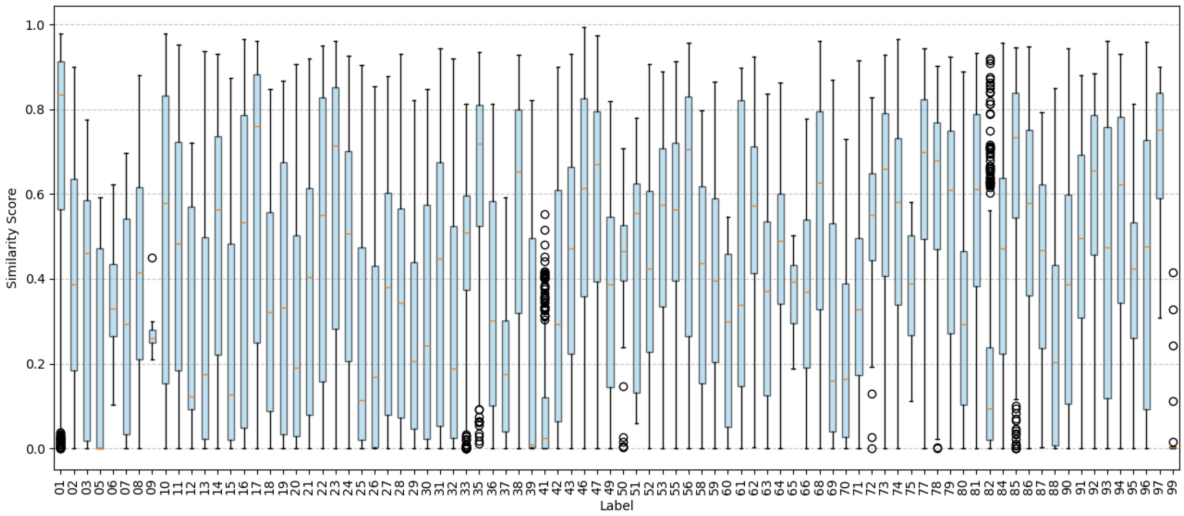
4.2.3 Conformal Risk Control

As for Conformal Risk Control, LDA probabilities produce smaller average prediction sets compared to cosine similarity scores (6.02 labels) and no empty sets on the validation split, while still satistfying the FNR control guarantee (empirical coverage $= 0.91$, $\tau = 0.01$). As for cosine similarity scores, the label-stratified distribution of coverage (Figure) is very similar to the Split Conformal Prediction procedure.

5. Conclusion

In this study, we examined language modelling-based methods for automatically classifying user queries into an official statistical taxonomy (ATECO, Istat). We used generative LLMs to create a synthetic dataset of labelled user queries and non-generative LLMs to encode these queries as embedding vectors in a high-dimensional semantic space. This allowed us to classify new queries using various strategies, such as cosine similarity-based retrieval and linear discriminant analysis. When evaluating our approach, we found relatively high agreement with the rule-based CIRCE method, exceeding 90% across the top five results. To set calibrated thresholds on cosine similarities and LDA output probabilities, and to extract prediction sets for each query with statistically guaranteed coverage, we employed Conformal Prediction (CP) techniques, namely split CP and Adaptive Prediction Sets (APS), as well as Conformal Risk Control using the false negative rate (FNR) as the loss function. We analysed the results in each scenario and found that APS procedures tend to yield better

Figure 4.5 - Distribution of output probabilities assigned to correct labels across ATECO divisions.



results as they produce smaller prediction sets while still guaranteeing marginal coverage.

Although additional work is needed, especially to develop retrieval strategies and fit discriminant models, these preliminary results are encouraging and suggest that the cumbersome process of defining deterministic rule sets can be simplified by using language models. This can greatly improve efficiency and scalability without sacrificing statistical robustness, thanks to modern model-agnostic and distribution-free uncertainty quantification techniques.

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Figure 4.6 - Average coverage for the split CP procedure (left axis) by ATECO division, ordered by normalised count (right axis). LDA probabilities as conformity scores.

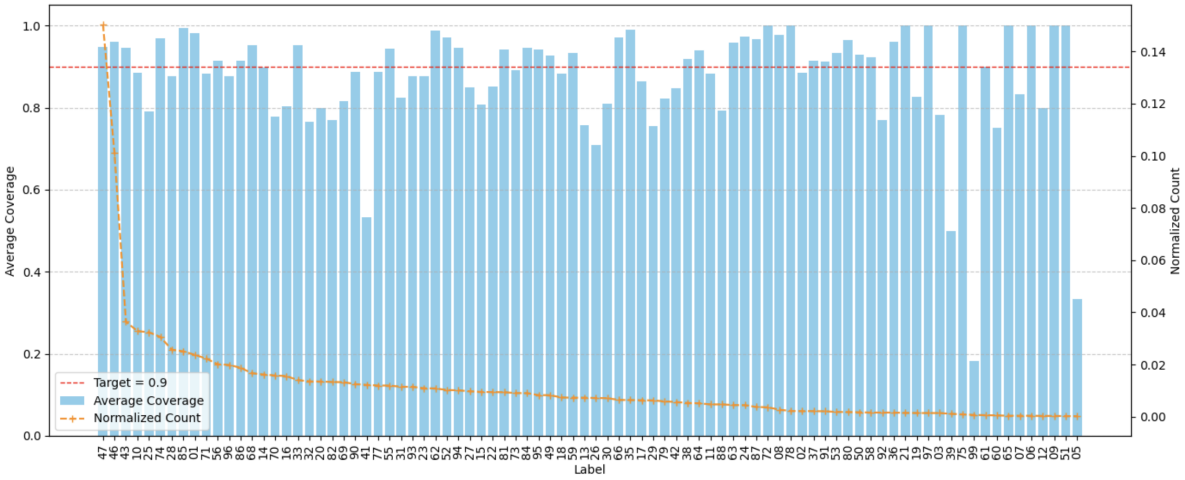

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Figure 4.7 - Average coverage for the APS procedure (left axis) by ATECO division, ordered by normalised count (right axis). LDA probabilities as conformity scores.

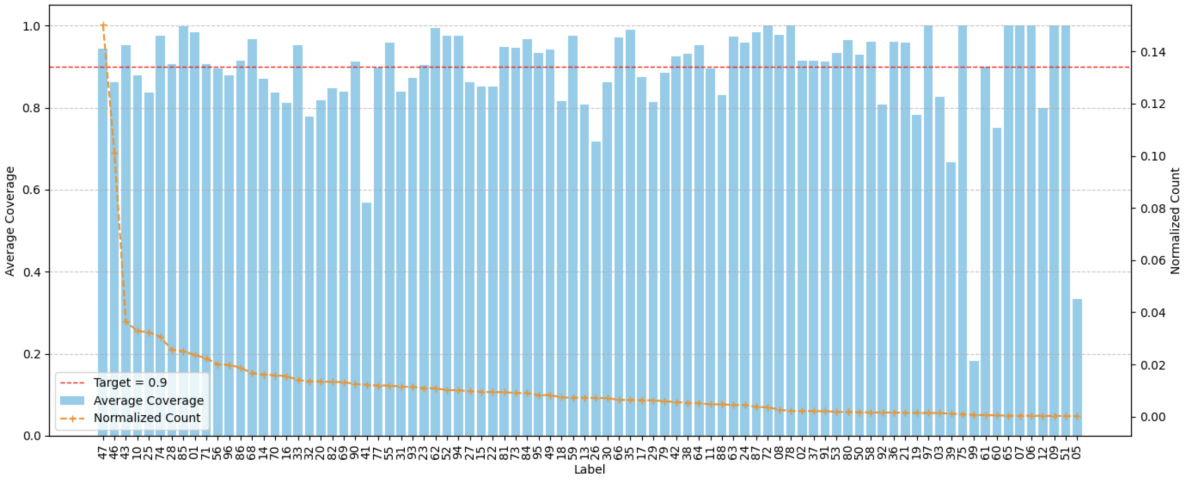

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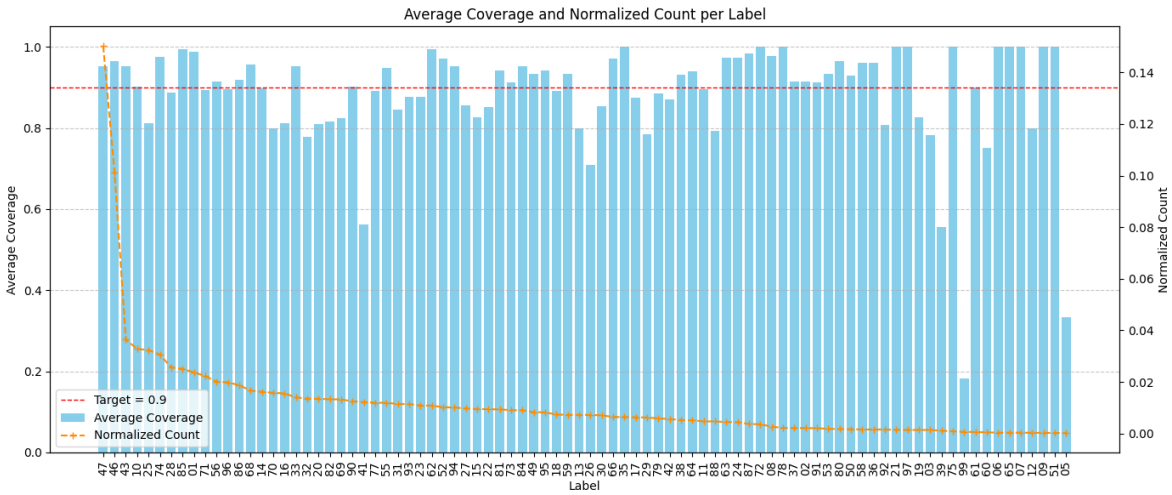
Figure 4.8 - Average coverage for the Conformal Risk Control procedure (left axis) by ATECO division, ordered by normalised count (right axis). LDA probabilities as conformity scores.



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